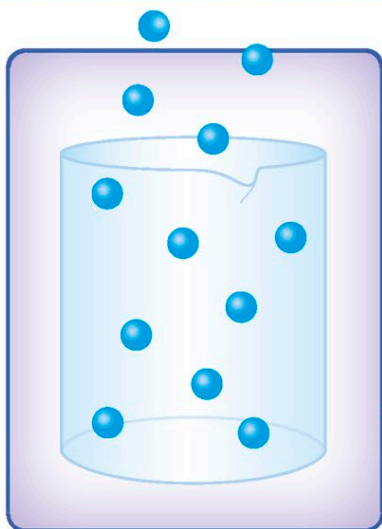


**Gas state**

The particles of a gas, such as oxygen, or the water vapor in the polar bear's breath, are not tightly held together by bonds. Without these forces keeping them together, they move freely in any direction.

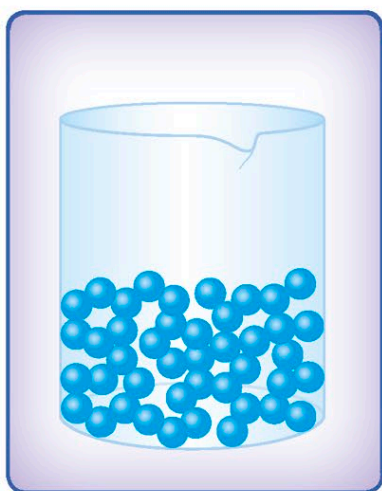
Air

Air is a mixture of gases: mostly nitrogen (78 percent), oxygen (21 percent), and small proportions of argon and carbon dioxide.

Solids, liquids, and gases

There are four different states of matter: solid, liquid, gas, and plasma. Everything in the universe is in one of those states. States can change depending on temperature and pressure.

All pure substances can exist in all of the three states common on Earth—solid, liquid, and gas. What state a substance is in is determined by how tightly its particles (atoms or molecules) are bound together. When energy (heat) is added, the tightly packed particles in a solid increase their vibration. With enough heat, they start moving around and the solid becomes a liquid. At boiling point, molecules start moving all over the place and the liquid becomes gas. Plasma is a type of gas so hot that its atoms split apart.

**Liquid state**

The particles of a liquid, such as water, are less tightly packed than in a solid and not neatly arranged, and they have weaker bonds. That is why liquids flow and spread, taking the shape of any container.

Salt water

Salty seawater has a lower freezing point than freshwater, which freezes at 32°F (0°C). Because salt disrupts the bonds between water molecules, seawater stays liquid until about 28°F (-2°C).

